Reading the grid

13 October 2026

The grid measures unconditional power: a changed protein counts in the denominator even when missingness prevents it from being tested. A large effect can make the test reliable for the proteins it can test while leaving total power below the target.

Testability caps twofold power

At n=12 samples per condition and m=64 cells per sample, k=1, the following estimates come directly from the frozen M3 table:

Preset	Unconditional power	Testability	Power / testability
specht2019v3	0.589	0.629	0.94
leduc2022_pSCoPE	0.609	0.647	0.94
derks2022_timsTOF	0.927	0.940	0.99
derks2022_QExactive	0.722	0.730	0.99
brunner2022	0.796	0.782	1.02

The ratios span 0.94 to 1.02 after rounding. For carrier TMT, pooled-carrier p_g and q_g do not respond to the injected effect; proteins failing the three-cell minimum in any sample remain untestable. At twofold and k=1, missingness therefore sets the main limit. Power / overall testability is a diagnostic ratio, not the new conditional-power column: power uses changed proteins and testability uses all proteins. Random changed-protein selection explains a ratio slightly above one. The new conditional power divides detected changed proteins by testable changed proteins within each replicate and cannot exceed one.

Donor variance removes 1.25-fold changes from this grid

No registered n, m or k combination reaches 80% mean power for a 1.25-fold change under the fixed Walzik donor variances. Increasing cells reduces cell-level noise and can improve testability; it does not remove the independent donor effect. More biological samples or smaller donor variance would define a different design or assumption and were not evaluated here. The grid cannot establish that whole-PBMC donor variance transfers to these cell lines; M4 injects the same external variance and cannot validate its transfer.

Read the k sensitivity alongside testability

At the same n=12, m=64 and twofold effect:

Carrier-TMT preset	k	Power	Testability
specht2019v3	1.00	0.589	0.629
specht2019v3	0.75	0.480	0.629
specht2019v3	0.50	0.217	0.628
leduc2022_pSCoPE	1.00	0.609	0.647
leduc2022_pSCoPE	0.75	0.521	0.647
leduc2022_pSCoPE	0.50	0.291	0.647

Compression leaves TMT detection assumptions unchanged but reduces the tested condition effect. Testability stays nearly constant while power falls as k moves from 1 to 0.75 or 0.5. This is a registered sensitivity analysis; k was not estimated and M4 cannot validate it. Read low realised FDR alongside mean rejection counts and the no-rejection fraction: a near-zero FDP can describe very few discoveries.

Software: R=4.6.1; scp=1.22.0; limma=3.68.5; edgeR=4.10.5; scplainer=workflow in scp 1.22.0; no separate package

This section is an interpretation amendment only. All original M3 numeric tables and plots remain unchanged; the original report is archived in power-report-v0.2.0.md. The original M3 run-manifest records that report snapshot; reading-grid-manifest.json records this addition.